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Spatio-temporal and time series analysis of bluetongue outbreaks with environmental factors extracted from Google Earth Engine (GEE) in Andhra Pradesh, India

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Abstract

This study describes the spatial and temporal patterns of bluetongue (BT) outbreaks with environmental factors in undivided Andhra Pradesh, India. Descriptive analysis of the reported BT outbreaks ($n = 2,697$) in the study period (2000–2017) revealed a higher frequency of outbreaks during monsoon and post-monsoon months. Correlation analysis of Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), rainfall and relative humidity (RH) displayed a significant positive correlation with BT outbreaks ($p < .05$). Retrospective unadjusted space–time, adjusted temporal and spatial analysis detected two, five and two statistically significant ($p < .05$) clusters, respectively. Time series distribution lag analysis examined the temporal patterns of BT outbreaks with environmental, biophysical factors and estimated that a decrease in 1 unit of rainfall (mm) was associated with 0.2% increase in the outbreak at lag 12 months. Similarly, a 1°C increase in land surface temperature (LST) was associated with 6.54% increase in the outbreaks at lag 12 months. However, an increase in 1 unit of wind speed (m/s) was associated with a 16% decrease in the outbreak at lag 10 months. The predictive model indicated that the peak of BT outbreaks were from October to December, the post-monsoon season in Andhra Pradesh region. The findings suggest that environmental factors influence BT outbreaks, and due to changes in climatic conditions, we may notice higher numbers of BT outbreaks in the coming years. The knowledge of spatial and temporal clustering of BT outbreaks may assist in adopting proper measures to prevent and control the BT spread.

KEYWORDS

bluetongue, clusters, google earth engine, season, sheep, time series analysis

1 | INTRODUCTION

The spatio-temporal patterns of vector-borne diseases are mostly driven by environmental factors which can be easily retrieved by

remote sensing (RS) and geographical information system (GIS) techniques. Geographical information system (GIS) and satellite-derived variables are highly useful in studying the disease distribution on an area-wide basis (Vinodh Kumar et al., 2016). Satellite-derived environmental variables such as vegetation indices, soil temperature and surface water were used in modelling the occurrence of

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infectious and vector-borne diseases (Baylis & Rawlings, 1998; Purse et al., 2004; Alkhamis and VanderWaal, 2016; Gao and Cao, 2019; Fekede et al., 2019; Liu et al., 2020). Besides meteorological parameters like temperature and rainfall, remote sensing variables are also relevant for estimating vector survival rates (Purse et al., 2005). Though studies on forecasting diseases based on environmental factors are emerging all over the world, the collection of remote sensing data is difficult, time-consuming, and high data storage capacity and specialized softwares are needed (Carpenter et al., 2009; Christaki, 2015). However, GEE is an advanced open-source geospatial platform based on JavaScript API with ease of computation, storage and cloud-based processing. It combines a multi-petabyte catalog of satellite imagery and geospatial data sets with planetary-scale analysis and makes it available for the researchers to apply in different fields (Climate Engine, 2016; Gorelick et al., 2017; Mutanga & Kumar, 2019).

Bluetongue (BT) is an infectious and non-contagious vector-borne viral disease that affects mainly small ruminants, especially sheep. BT is caused by Bluetongue virus (BTV), belongs to *Orbivirus* genus in the *Reoviridae* family, which is transmitted between ruminant hosts by blood-sucking midges of the *Culicoides* spp (Chand et al., 2015; MacLachlan, 2011). The spatial distribution of *Culicoides* species is limited by several environmental factors, including temperature, soil properties and land cover (Carpenter et al., 2009). Currently, there are 28 characterized serotypes of BTV exists worldwide (Bumbarov et al., 2020). BT causes economic losses, mainly due to high morbidity, mortality, abortions, loss of meat and fleece. Morbidity in sheep can reach 100% with mortality between 30% and 70% in more susceptible breeds (Singh & Shive, 2009). In addition, the presence of BTV has become a major impediment to the international trade of ruminants and their products. The virus has spread widely in the tropics and subtropics, approximately between the latitudes 35°S and 40°N (Mellor et al., 2000). In recent years, the effects of climate change have contributed to an extension of its distribution area into higher latitudes and facilitated major outbreaks in Europe (Mellor et al., 2008; Saegerman et al., 2008). In India, BT outbreak was first reported in 1960s, since then it is endemic in India; however, the occurrence of BT in India varies based on the time of rainfall, temperature, moisture conditions, sheep population and susceptible sheep breeds (Prasad et al., 1992, 2009; Ranjan et al., 2015; Rupner et al., 2020; Sapre, 1964; Sreenivasulu et al., 2004). Overt clinical outbreaks have mostly been reported from rural native mixed flocks of sheep and goats across the four states of Telangana, Andhra Pradesh, Karnataka and Tamil Nadu in the southern region of India (Babu et al., 1988; Khan et al., 1983; Rao et al., 2016).

In recent years, spatial epidemiologists successfully modelled BT outbreaks and distribution of its vector in Europe using remote sensing and GIS (Cuéllar et al., 2020; Samy & Peterson, 2016). Studies used satellite data to identify the environmental risk factors for the emergence of BT in Europe and Asian countries (Brand & Keeling, 2017; Carpenter et al., 2009; Chanda et al., 2019; Gao

et al., 2017; Mellor & Wittmann, 2002). Understanding the pattern of occurrence of BT will help the animal husbandry authorities to make preparedness and control. Hence, this study aims to investigate the spatio-temporal clustering and the effect of environmental variables on the BT outbreaks in undivided Andhra Pradesh, India.

2 | MATERIALS AND METHODS

2.1 | Study area

The study was performed in undivided Andhra Pradesh (AP), geographically lies between 12°37'N - 19°54'N latitudes and 76°45'E - 84°46'E longitudes in the southern part of India. However, the state was divided into two parts (Telangana and Andhra Pradesh) in the year of 2014. Andhra Pradesh (26.39 million) ranks first in the total sheep population (65.06 million) of India (India, 19th Livestock Census 2012), and sheep are reared in the fluvial deltas, plains, mountain ranges, sandy tracts, highlands and also in the plateaus of the state (Johnson et al., 2006). There are four major seasons in AP: summer (March–May), monsoon (rainy) (June–September), post-monsoon (October–December) and winter (January–February). The state receives rainfall from both South-West (67%) and North-East monsoon (25%). The spatial variation of temperature from one part of state to the other is not much significant except in specific tiny patches at high altitude. The maximum temperature is recorded in summer season varying between 37° and 44°C, and the minimum temperatures in winter season ranging between 14° and 19°C (Meteorological Center, Hyderabad).

2.2 | Data collection and processing

Epidemiological data on BT outbreaks in AP from 2000 to 2017 were collected from the Department of Animal Husbandry, Government of Telangana. The district-wise sheep population data were collected from three livestock censuses of the Department of Animal Husbandry, Dairying and Fisheries (DAHDF), India (17th Livestock Census [2000–05], 18th Livestock Census [2006–10], 19th census [2011–17]). The environmental variables were collected using Google Earth Engine (GEE), which is an intermediary platform for remote sensing data sets. In GEE, preprocessing of required data like filtration for cloud mask, water mask, filtration date and boundaries for each district has been carried out (Huang et al., 2017; Jamei et al., 2019; Mateo-García et al., 2018). After filtration, the data were exported as comma-separated values (CSV) files in Google Drive. Seven environmental variables such as temperature (°C), rainfall (mm), Wind Speed (WS) (m/s), Relative Humidity (RH) (%), Normalized Difference Vegetation Index (NDVI), Land Surface Temperature (LST) and Normalized Difference Water Index (NDWI) were collected from sources like Moderate Resolution Imaging Spectroradiometer (MODIS), National Center for Environmental Prediction Reanalysis Information (NCEP_RE), Tropical Rainfall Measuring Mission

TABLE 1 Details of the environmental data and the satellite sources collected from Google Earth Engine

Variables (Monthly)	Period	Source	Resolution	Cadence	References
Normalized difference vegetation index	2000–2017	MODIS/006/MOD13Q1	250 meters	Daily	Didan, 2015
Land surface temperature	2000–2017	MODIS/006/MOD11A1	1,000 meters	Daily	Wan et al., 2015
Normalized difference water index	2000–2017	MODIS/MOD09GA_NDWI	250 meters	Daily	Vermote & Wolfe, 2015
Temperature	2000–2017	NCEP_RE/surface temp	2.5 arc degrees	6 hr	Kalnay et al., 1996
Precipitation	2000–2017	TRMM/3B42	0.25 arc degrees	3 hr	Huffman et al., 1995
Wind speed	2000–2017	NASA/GLDAS/V021/NOAH/G025/T3H	0.25 arc degrees	3 hr	Rodell et al., 2004
Relative humidity	2000–2017	NASA/GLDAS/V20/NOAH/G025/T3H	0.25 arc degrees	3 hr	Rodell et al., 2004

(TRMM) and Global Land Data Assimilation System (GLDAS) by writing JavaScript codes (Table 1). Spatial maps were generated with ArcGIS 10.3 (ESRI 2014. ArcGIS Desktop: Release 10.3. Redlands, CA: Environmental Systems Research Institute).

2.3 | Descriptive and inferential analysis

All the data collected from relevant sources were curated, and descriptive analysis was carried out. The correlation between BT outbreaks and environmental variables was plotted with significance ($p < .05$) in R statistical software 4.0.2 (R core team, 2020).

2.4 | Spatio-temporal cluster analysis

A retrospective space–time statistical scan was performed to identify the presence of high-risk space–time clusters of BT outbreaks over the study period using SaTScan v9.6 (Kulldorff et al., 1998). A cylindrical window with a circular geographical base and the height corresponding to time was used to scan through space and time statistics (Pruthivishree et al., 2017). The possible spatio-temporal clusters were detected by the dynamic flux of base and height of the cylindrical window. Log-likelihood ratio (LLR) was used to evaluate the risk within and outside the window, and scanning window with the highest LLR was defined as the most-likelihood cluster. This study analysed spatial and temporal adjustments, including (a) unadjusted space–time Poisson discrete model, (b) space–time analysis with spatial adjustment using non-parametric, spatially stratified randomization, (c) space–time analysis with temporal trend adjustment using non-parametric, time stratified randomization. The adjustments were made to find the influence of spatial and temporal trends (Kulldorff, 2006). For all scan tests, the centroid of each district's latitude and longitude was used to create the coordinates file. A maximum spatial cluster size of 50% of the population at risk and maximum temporal cluster size of 50% of the study period were used. For scan tests, 999 Monte Carlo replications were used to estimate the significance levels of the clusters. For each analysis, the most likely (based on the size of the log-likelihood ratio), non-overlapping statistically significant ($\alpha = 0.05$) clusters are presented.

2.5 | Time series distributed lag analysis

Usually, time series regression is used to examine the dependent variable (Y) with the independent variables ($X_1, X_2, \dots$). We must consider when it has the effect (time) instead of how much effect independent variables (X_t ; Environmental factors) have on the dependent variable (Y_t ; BT outbreak). To understand this, we must estimate the lag distribution relating Y_t to X_t . So Distributed Lag Model (DLM) explains the effect of an independent variable X on a dependent variable Y occurs over time (t). Therefore, DLMs are dynamic models,

and a finite linear DLM with one independent variable is written as follows (Baltagi, 2008; Hill et al., 2001).

$$Y_t = \alpha + \sum \beta_s X_{t-s} + e_t \quad (1)$$

$$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \beta_2 X_{t-2} + \dots + \beta_n X_{t-n} + e_t \quad (2)$$

where Y_t denotes the t^{th} observation on the dependent variable Y , and X_{t-s} denotes the $(t-s)^{\text{th}}$ observation on the independent variable X . α is the intercept, and $\beta_0, \beta_1, \dots, \beta_s$ are the current and lagged coefficients of X_t . The parameter set is the stationary error term for the time point t with zero mean and constant variance.

$$(E(e_t) = 0, \text{var}(e_t) = \sigma^2, \text{and } \text{cov}(e_t, e_s) = 0).$$

The equation of finite linear DLM with multiple independent variables has the form.

$$Y_t = \alpha + \sum \beta_s \text{RF}_{t-s} + \sum \beta_s \text{WS}_{t-s} + \sum \beta_s \text{LST}_{t-s} + e_t \quad (3)$$

where Y is the monthly count of BT outbreak, RF is the average rainfall at lag ' s ' month, WS is the average wind speed at lag ' s ' month, and LST is the average land surface temperature at the lag ' s ' month. The monthly average rainfall (mm), wind speed (m/s) and land surface temperature ($^{\circ}\text{C}$) were selected as the reference to estimate the risk of BT outbreak associated with each unit increase, respectively. The best prediction model was selected based on the lowest Akaike's information criteria (AIC) and Bayesian information criteria (BIC). The optimal model was used to predict the BT outbreak for 12 months of the year. All the analyses were done by using the 'dLagM' package in R 4.0.2 (Demirhan, 2020).

3 | RESULTS

3.1 | Descriptive analysis

Data on BT outbreaks in AP from 2000–2017 showed variation in yearly occurrence. A total of 2,697 outbreaks were reported during the study period (Table S1) and the highest number of outbreaks was reported in 2005. However, no outbreaks were reported in 2011, 2014 and 2015. The majority of outbreaks (98%) were reported in the post-monsoon (57%, $n = 1,552$) season followed by monsoon season (41%, $n = 1,094$) (Figure 1).

Environmental variables were extracted from the GEE (2000–2017) because of its ease of computation, storage and cloud-based processing. Andhra Pradesh receives rainfall from both North-West monsoon and South-East monsoon from June to December. During the study period, more than 80% of the rainfall received between June and October. The NDVI and LST ranged between 0.26 and 0.65 and 26° and 42°C , respectively. Monthly NDWI ranged between -0.11 and 0.08 , and temperature ranged between 20° and 34°C . The WS ranged between 1.8 and 4.43 m/s, with an average of 2.71 m/s. The RH range was 31%–86%, with an average of 60% (Table 2).

3.2 | Correlation of BT outbreaks with environmental factors

Correlation analysis showed that the variables like NDVI ($r = .32$), NDWI ($r = .27$), rainfall ($r = .23$) and RH ($r = .31$) showed a significant

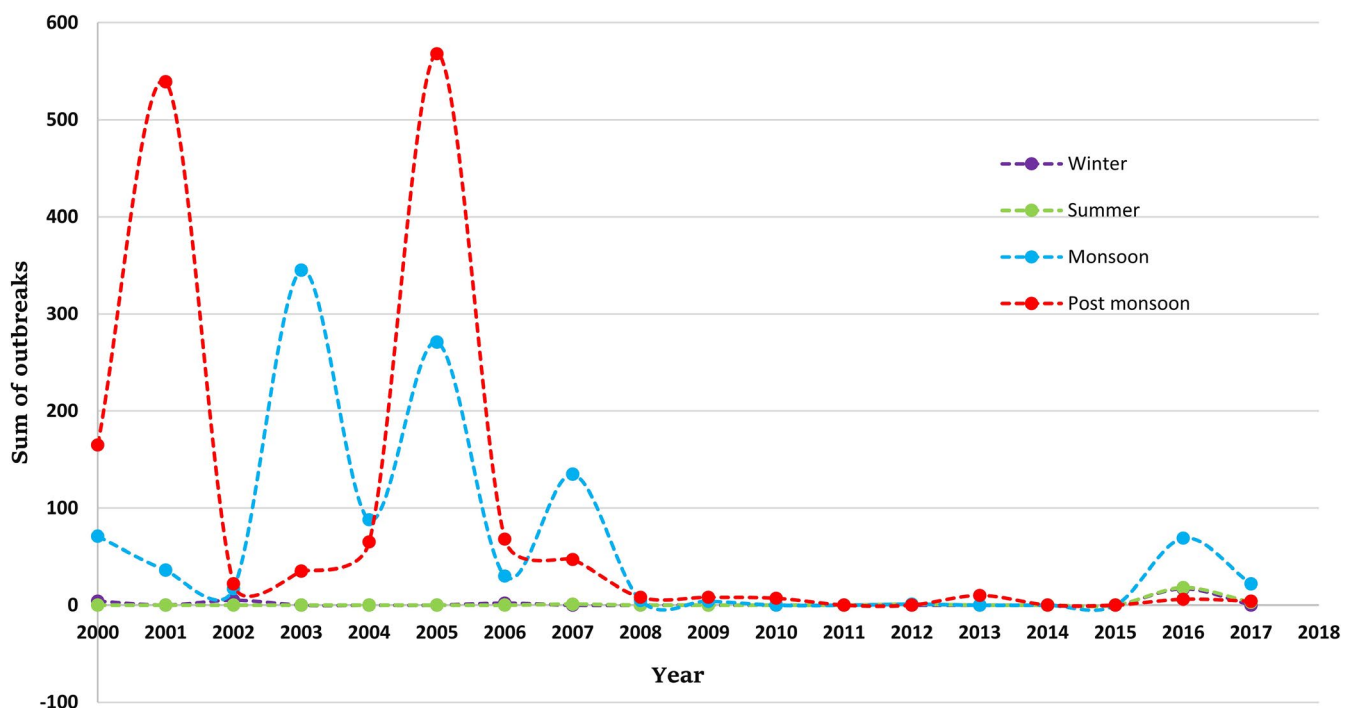


FIGURE 1 Year and season wise number of BT outbreaks in undivided Andhra Pradesh

TABLE 2 Descriptive statistics of environmental variables for Andhra Pradesh

Variables	Mean (SD)	Minimum	Maximum
Temperature (°C)	26.63 (3.56)	20.97	34.86
Cumulative rainfall (mm)	75.54 (76.07)	0.00	338.00
Windspeed (m/s)	2.71 (0.54)	1.80	4.43
RH (%)	60.47 (14.75)	31.00	86.76
NDVI	0.46 (0.10)	0.26	0.65
LST (°C)	33.17 (4.15)	26.95	42.00
NDWI	-0.03 (0.05)	-0.11	0.08

positive correlation, ($p < .05$) while WS ($r = -0.16$), temperature ($r = -0.17$) and LST ($r = -0.22$) showed a significant ($p < .05$) negative correlation (Figure 2).

3.3 | Space-time cluster analysis

The spatio-temporal clusters of BT outbreaks are presented in Figure 3 & Table 3. The unadjusted, spatial and temporal adjusted models showed statistically significant ($p < .01$) single and multi-centre clusters during the study period.

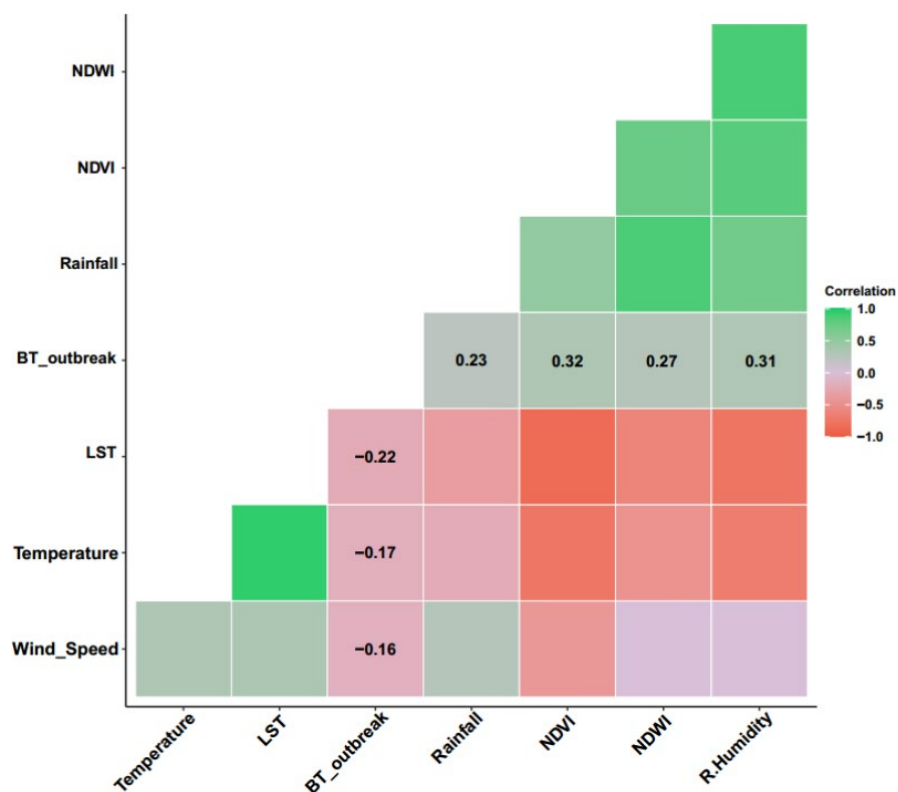
The unadjusted space-time analysis with 50% of the population at risk detected two statistically significant ($p < .01$) clusters (Figure 3). The first cluster with highest LLR was comprised of Anantapur, and the other multi-centre cluster was comprised of coastal Andhra Pradesh and part of the Telangana region (15 districts). The temporal

adjusted space-time analysis detected five statistically significant clusters ($p < .01$) with highest LLR in a single-centre cluster comprised of the Anantapur district. The spatial adjusted space-time analysis detected two statistically significant clusters ($p < .01$) with highest LLR in Anantapur, and the second multi-centre cluster was detected at the inner Andhra Pradesh region (9 districts). In all the three-cluster analysis, Anantapur district was identified as a statistically significant ($p < .01$) cluster (Table 2).

3.4 | Temporal patterns of BT outbreaks

Overall, 2,697 BT outbreaks were recorded during the study period (2000–2017) in AP. In particular, most (99%) of the outbreaks have occurred in the months of post-monsoon (60%) and monsoon (39%) months, respectively. The distributed lag model (DLM) revealed that rainfall and LST at lag 12th month, WS at lag 10th month have a statistically significant association ($p < .05$) with BT outbreaks. The DLM results revealed a negative relationship with rainfall at lag 12 months. A decrease in 1 unit of rainfall (mm) was associated with a 0.2% increase in the outbreak at lag 12 months. An increase in 1 unit of WS (m/s) was also associated with a 16% decrease in the outbreak at lag 10 months. Similarly, a 1°C increase in LST was associated with a 6.54% increase in the outbreak at lag 12 months (Table 4).

The smallest AIC and BIC values have identified the optimal multivariable model and the best model comprised of rainfall at lag 12 months, wind speed at lag 10 months and land surface temperature at lag 12 months (Table 5). The temporal pattern of BT outbreaks predicted from the optimal model is presented in Figure 4. It

**FIGURE 2** Correlogram of BT outbreaks with environmental factors (Significant at 5% level [$p < .05$])

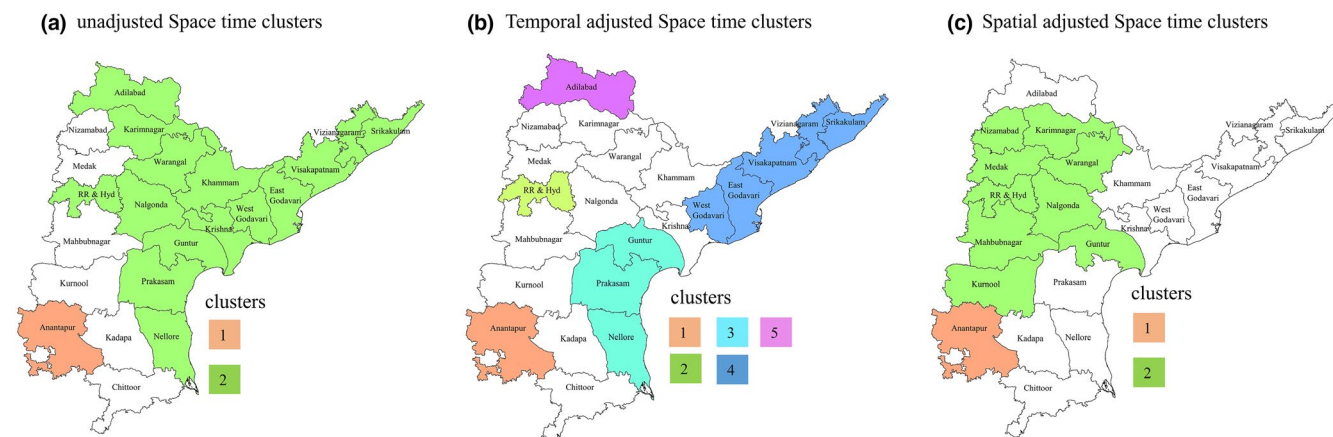


FIGURE 3 High rated significant spatial-temporal clusters of bluetongue outbreaks with maximum spatial window = 50% of population at risk and maximum temporal window = 50% of the entire study period. a) Unadjusted space-time clusters, b) temporal adjusted space-time clusters and c) Spatial adjusted space-time clusters

showed that BT outbreaks tend to increase from August and reach a peak in December, which is the post-monsoon season in AP.

4 | DISCUSSION

This study detailed the relationship of BT outbreaks with environmental factors, spatio-temporal distribution and time series distributed lag model of BT in AP. The environmental variables were preprocessed and downloaded via GEE platform from various sources like MODIS, NCEP_RE, TRMM and GLDAS. Traditionally, remote sensing and meteorological variables were collected from satellites in a low-Earth orbit (LEO) like Landsat, MODIS, Palsar and Sentinel, nominally altitudes between 300 and 1,500 km (Ackerman et al., 2018). Organizations like National Oceanic and Atmospheric Administration (NOAA) made free availability of these satellite images (Hare et al., 2004). But this method consumes high computational power and time to process. Since its launch in 2010, GEE has been used for various studies at different scales. Because it is like the planet in our hand and includes more than forty years of historical imagery of almost all the satellites and scientific data sets, which can be processed and readily used with World of JavaScript coding. In this study, we collected the environmental variables from GEE platform by writing codes because of its advantage over other methods of collection. The data are continuous for 18 years of the study period, with high resolution and covered whole study area.

In AP, maximum number of BT outbreaks was recorded during September and October months, which is the transition period from monsoon to post-monsoon in the region. The probable reason for higher outbreaks in this period may be favourable environmental conditions like moderate rainfall (50–150 mm/month), humidity (69%–76%) and vegetation (0.21–0.27), which makes the suitable condition for vector multiplication. A similar kind of association has been reported in other BT outbreaks (Braverman et al., 2001; Subramanian & Piramanayagam, 2001). In this study, the positive relationship between NDVI, NDWI and rainfall showed the importance

of its effect on life history of biting insects. During the study period, less or no outbreaks were recorded between March and May, and it may be associated with dry weather due to temperature (36°–40°C), minimal vegetation (0.05–0.1) and less vector population (Reddy & Hafeez, 2008). While in post-monsoon, higher outbreaks were observed, which is similar in other South Indian states like Tamil Nadu (Selvaraju et al., 2013). The influence of temperature on BT outbreaks has been studied extensively in the European region, and moderate temperature favours BT transmission dynamics (Brand & Keeling, 2017; Guis et al., 2012). As a consequence of global climate change, the distribution and nature of BTV infection have changed significantly in current years (Purse et al., 2005; Samy & Peterson, 2016). Recently, it has been estimated that by 2,100, the geographical risk of BT outbreaks may extend further north and also increase in the seasonal transmission with an average increase in outbreaks (Jones et al., 2019).

Space-time analysis was aimed at testing for the presence of space-time interaction, or the presence of clusters of BT outbreaks over the study period. Stojmanovski & Mayakovski (2018) conducted a similar kind of study in the Balkan peninsula and identified the significant hotspot areas for BT. The cluster analysis performed in this study showed high rated clusters with no overlapping, and most of the clusters were formed during 2000–2005, where 80% of the outbreaks were reported. It suggested that disease transmission may be different in the diverse areas, perhaps involving different vectors or different hosts and different environments for transmission, or any combination of these effects corroborates the findings of Bouchemla et al. (2017). The three types of spatio-temporal cluster analysis detected Anantapur district as a statistically significant single-centre cluster, and the probable reason may be due to the highest sheep population and density in that district.

The study depicted districts inside clusters had an increasing number of outbreaks compared with outside districts. Interactions are said to occur when disease cases occur closer together in either or both of space and time that would be the situation without interaction when cases would occur at random in both space and

TABLE 3 High rate spatial clusters with 50% population at risk in Space-time Poisson discrete scan statistics

Clusters	Timeframe	Districts included	Number of outbreaks	Expected outbreaks	Annual cases/100,000	Relative risk	Log likelihood ratio (LLR)	p-value
Space-time clusters								
1	2001/1–2001/12	Anantapur	450	14.37	23.6	37.8	1,153.99	<.01
2	2003/1–2005/12	East Godavari, West Godavari, Visakapatnam, Krishna, Khammam, Vizianagaram, Guntur, Warangal, Srikakulam, Nalgonda, Prakasam, Karimnagar, Nellore, Adilabad, RR&H&Yd	1,016	259.03	3	5.85	772.69	<.01
Space-time clusters – Spatial adjustment								
1	2001/1–2001/12	Anantapur	450	24.56	13.8	22.02	921.2	<.01
2	2005/1–2005/12	Rangareddy, Hyderabad, Medak, Mahabubnagar, Nalgonda, Nizamabad, Karimnagar, Warangal, Kurnool, Guntur	535	61.86	6.5	10.67	729.07	<.01
Space-time clusters – Temporal adjustment								
1	2001/1–2001/12	Anantapur	450	51.27	6.6	10.44	612.27	<.01
2	2005/1–2005/12	Rangareddy and Hyderabad	215	23.33	7	9.97	293.25	<.01
3	2002/1–2008/12	Prakasam, Guntur, Nellore	529	209.65	1.9	2.92	193.03	<.01
4	2003/1–2003/12	Srikakulam, Vizianagaram, Visakapatnam, East Godavari, West Godavari, Khammam, Krishna	126	40.77	2.3	3.2	58.4	<.01
5	2003/1–2003/12	Adilabad	51	10.39	3.7	4.99	40.87	<.01

TABLE 4 The lag effects of environmental factors on BT outbreaks (distributed lag model)

	Rainfall (mm)			Wind speed (m/s)			LST (°C)		
lag (months)	Change (%)	SE (LL-UL)		Change (%)	SE (LL-UL)		Change (%)	SE (LL-UL)	
0	0.16	0.07	0.25	8.17	0.57	15.77	-0.62	-3.82	2.58
1	0.18	0.09	0.27	2.39	-5.21	9.99	-2.56	-6.06	0.94
2	0.09	0	0.18	-4.29	-11.79	3.21	1.14	-2.26	4.54
3	0.1	0.01	0.19	-0.39	-7.89	7.11	0.6	-2.9	4.1
4	-0.18	-0.27	-0.09	5.08	-2.42	12.58	3.87	0.37	7.37
5	-0.03	-0.12	0.06	-6.53	-14.13	1.07	2.77	-0.63	6.17
6	-0.02	-0.11	0.07	-2.66	-10.36	5.04	1.81	-1.69	5.31
7	0.04	-0.05	0.13	-7.89	-15.59	-0.19	3.88	0.38	7.38
8	0.01	-0.08	0.1	-5.11	-12.81	2.59	-0.27	-3.77	3.23
9	-0.02	-0.11	0.07	0.92	-6.68	8.52	2.01	-1.49	5.51
10	-0.02	-0.11	0.07	-16.07 [*]	-23.67	-8.47	-0.18	-3.58	3.22
11	-0.05	-0.14	0.04	-14.94	-22.84	-7.04	0.4	-3.1	3.9
12	-0.20 [*]	-0.29	-0.11	-8.76	-16.66	-0.86	6.54 [*]	3.34	9.74

Note: SE = Standard Error (LL = Lower Limit–Upper Limit).

*Significant at 5% level ($p < .05$).

time (Hsu et al., 2007; Kulldorff et al., 1998). Both the temporal and spatial adjustments with stratified randomization identified the nearly similar clusters compared with the unadjusted clusters. Clustering and space–time interaction are not designed to quantify the relationship between the dependent and independent variables (Kulldorff, 2006). The spatial scan results showed that the centres of BT clusters were mostly located at the districts, which has the highest population density. Possibly, the higher population density and

TABLE 5 The lag effects of environmental factors on BT outbreaks (Multiple distributed lag model)

Model	% change (±SE)	AIC	BIC	p-value
Model-I ^a				
Lag 12: Rainfall	0.08 (0.06)	1,715.1	1,727.3	.003**
lag 10: WS	-16.5 (7.65)			
Lag 12: LST	3.11 (1.33)			
Model-II				
Lag 12: Rainfall	0.08 (0.06)	1,718.56	1,727.71	.014*
lag 10: WS	1.18 (1.07)			
Model-III				
Lag 12: Rainfall	0.06 (0.05)	1,717.78	1,728.93	.009*
Lag 12: LST	0.26 (0.18)			
Model-IV				
lag 10: WS	-15.36 (7.6)	1,715.03	1,724.18	.002**
Lag 12: LST	3.08 (1.34)			

Abbreviations: AIC, Akaike information criteria; BIC, Bayesian Information Criteria; SE, Standard Error.

^aThe best model has the lowest values of AIC and BIC.

*Significant at 5% level ($p < .05$).

**Significant at 0.5% level ($p < .005$).

high-frequency contact may help the BT viruses to spread (Chanda et al., 2019). The knowledge gained on the spatial clusters of the BT outbreaks not only oblige authorities to take targeted preventive measures within the clusters but also allow them to monitor a specific region with high-risk areas and periods.

All the remote sensing and climatic variables such as NDVI, LST, NDWI, temperature, wind speed, humidity and rainfall depicted statistically significant association with BT outbreaks. Earlier studies on spatial pattern of vector borne diseases used mean climatic values focused small geographical regions, limited to single vector and disease, have highlighted the need for expanded spatio-temporal analysis (Patz et al., 2005; Reiter, 2001).

There are few reports available on the use of DLM in infectious disease modelling such as the association of dengue outbreaks with temperature (Kakarla et al., 2019) and an integrated model that included distributed lags for temperature and precipitation accurately predicted the resurgence of the West Nile virus (WNV) in humans South Dakota, the USA in 2016 (Davis et al., 2017). This study also reported a predictive model for BT outbreaks by multiple DLM. The use of both dependent, independent environmental variable, and their lags was efficient than the dependent variable, and its lags since the independent environmental factors also play a role in the outbreak of vector-borne diseases (Bhaskaran et al., 2013). Ward and Thurmond (1995) reported the lag effect of temperature and rainfall on vector population dynamics of BT. The present study identified the lag effect influence of rainfall, WS and LST. The lack of association of other variables may be due to the modelled average patterns in environmental factors with BT outbreaks over a long time or may be due to applied spatial scale units. By averaging, the patterns across the state may also miss the local scale effect of environmental factors on the BTV system. Limited studies are available on BT depicting the quantitative links with environmental variables, accounting for the

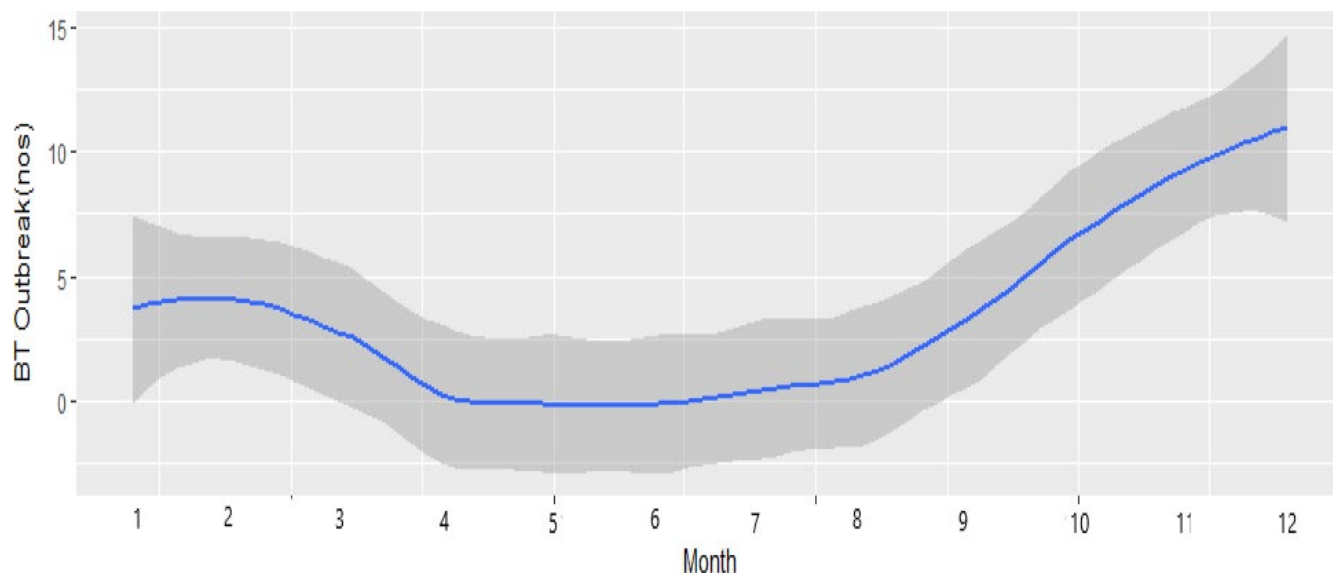


FIGURE 4 Temporal pattern of BT outbreak across 12 months of a year. The outbreak is predicted by model I in Table 2 (lag-12 rainfall, lag-10 wind speed [WS] and lag-12 Land Surface Temperature [LST])

temporal and spatial relationship of the dependent variable (Cazelles et al., 2007; Chanda et al., 2019; Chou et al., 2010). The lag relationship of rainfall suggesting that the outbreaks in the post-monsoon season may be due to the environmental suitability multi-vector systems from the monsoon rainfall. This effect of rainfall variability in the monsoon on the BT epidemics has been hypothesized previously also in the Indian subcontinent (Prasad et al., 2009). The role of wind speed on the BT outbreaks remains unquantified and still uncertain. But the dispersal of the *Culicoides* midges with the help of wind may influence the spatio-temporal occurrence of BT epidemics (Sanders et al., 2010; Sedda et al., 2012). But the present study showed the significant influence of the lag wind speed on the BT epidemics. The temperature effect on the BT virus itself or the vectors has been reported efficiently (Gao et al., 2017; Sellers & Mellor, 1993). However, LST is reported to influence the survivability of the vectors (Tatem et al., 2003).

The study reported the strong association of environmental factors like rainfall, WS and LST with BT outbreaks. However, other factors like temperature, RH, NDVI and NDWI were not shown a better association, but there is a correlation with the BT outbreaks. The knowledge of spatial and temporal clustering of BT outbreaks may assist in adopting proper measures to prevent and control the spread. Though the study identified spatio-temporal clusters and a better model for prediction of BT outbreaks, it has some limitations such as non-availability of the point data (outbreak villages) and details of the sheep breeds affected.

5 | CONCLUSION

The findings of the present study highlight the use of GEE as an intermediate platform for the collection of satellite-derived

environmental variables. It is the first of its kind in utilizing the climatic and remote sensing variables from the GEE platform for livestock disease in India. The time series analysis of BT outbreaks in AP with environmental factors highlights the importance of lagged climatic on BT outbreaks. A predictive model of BT outbreaks was created, and high-risk areas of BT outbreaks were determined, and these can support the development of an early warning system for BT outbreaks in AP. Further studies are needed about factors influencing BT, which focus on both ecological, behaviour factors, social and economic effects of BT, and prevention measures in AP to provide a more comprehensive understanding of the interaction between climate change-related factors and other factors.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Ravichandran Karthikeyan involved in data curation, formal analysis and writing—original draft. Ramkumar involved in methodology, data curation. Shiva Reddy involved in methodology, resources and software. J. Nagaraj involved in methodology, formal analysis and validation. D.K. Sinha, B.R Singh and Yashpal Singh Malik involved in validation. Vinodh Kumar O. R involved in conceptualization, data curation, formal analysis, writing—original draft and project administration.

ETHICAL APPROVAL

Not applicable.

DATA AVAILABILITY STATEMENT

I confirm that I have included a citation for available data in my references section, unless my article type is exempt.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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